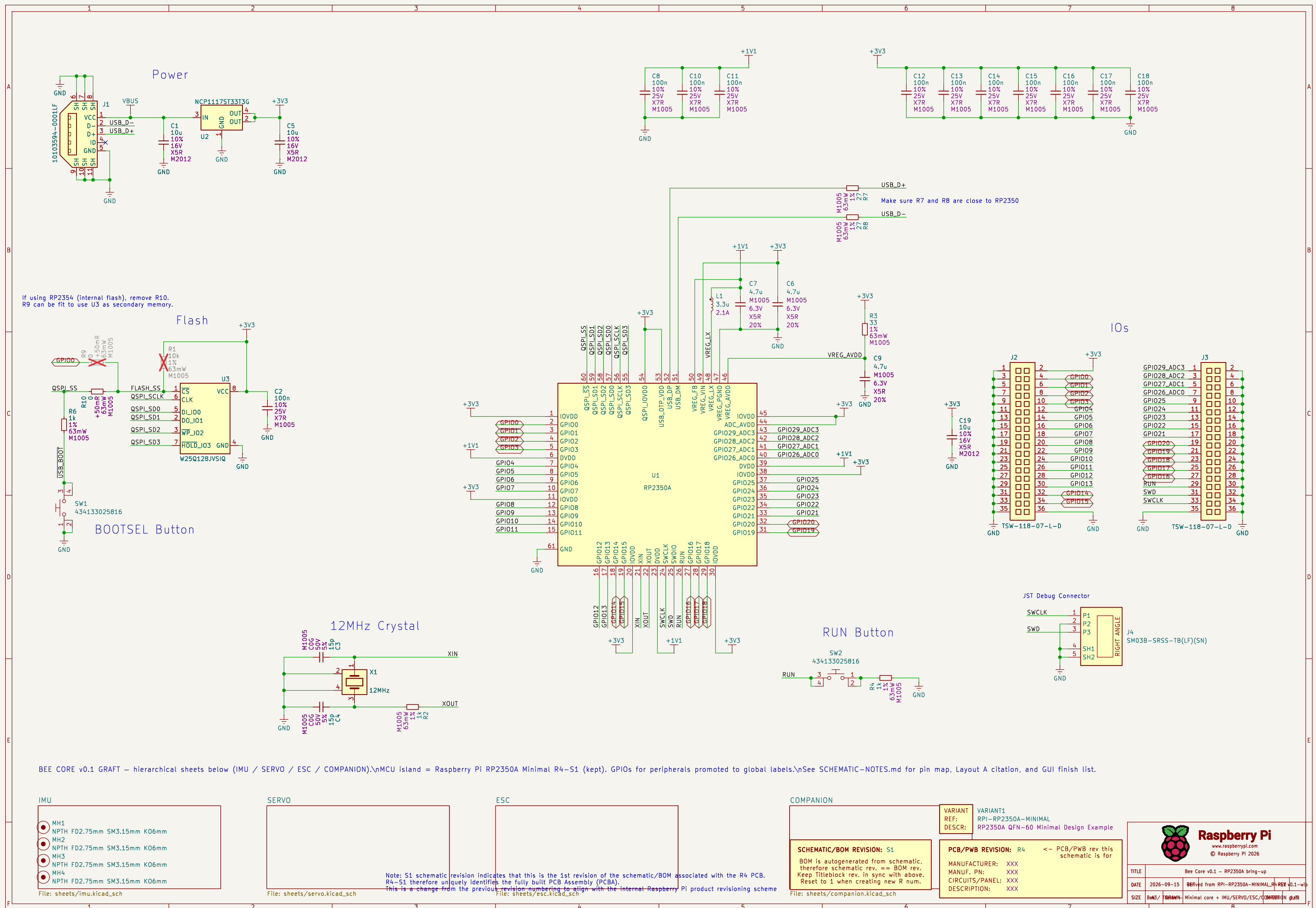
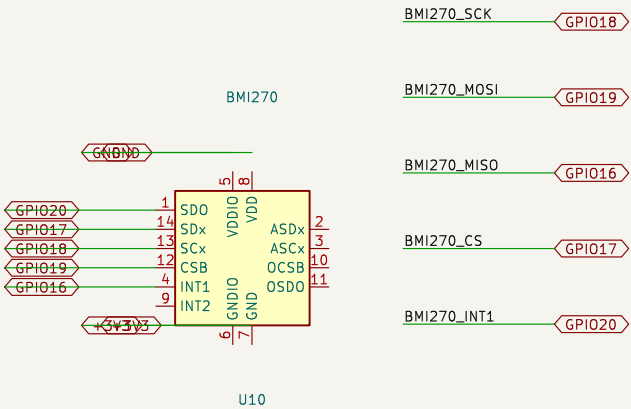




U? BMI270 pin map (datasheet):\n1 SDO=MISO <- GPIO16\n4 INT1 -> GPIO20\n5 VDDIO = +3V3\n6 GNDIO = GND\n7 GND = GND\n8 VDD = +3V3\n12 CSB <- GPIO17 (active low)\n13 SCx = SCK <- GPIO18\n14 SDx = MOSI <- GPIO19\n2,3,9,10,11 NC (no aux/OIS on v0.1)\n



TP: BMI270_CS, BMI270_INT1

SECTION: IMU – Bosch BMI270 (LCSC C2836813) on SPI0\nSPI: GPIO18=SCK, GPIO19=MOSI(SDx), GPIO16=MISO(SDO), GPIO17=CSB\nINT1=GPIO20. VDD=VDDIO=+3V3. Aux/OIS pins NC. Decoupling cap on near VDD/VDDIO.\nFootprint: Package



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TITLE	Bee Core – IMU (BMI270 SPI)				
DATE	2026-09-15	REF	Bee Core bring-up – graft sheet	REV	v0.1-wip
SIZE	A4	DRAWN	via global labels to Minimal RP2350	SHEET	2/5



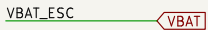
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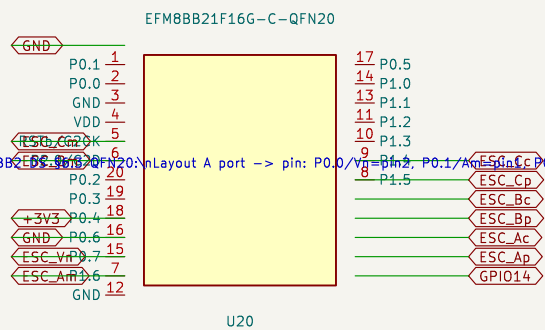
TITLE	Bee Core – SERVO PWM + 5V				
DATE	2026-09-15	REF	Bee Core bring-up – graft sheet	REV	v0.1-wip
SIZE	A4	DRAWN	via global labels to Minimal RP2350	SHEET	3/5

FLAG/GUI: place bee--core:EFM8BB21F16G--C-QFN20 + 3x A03401A + 3x A03400A + gate R +\nsense R network + VBAT pad/polyfuse + bulk C. Prefer correct nets over fake completeness.\nUNCERTAIN: P-FET full enhancement at Vgs=-(4.2-3.3)= -0.9V on 1S is NOT enough --\nwhoop designs often gate P-FETs from VBAT--referenced drivers or use different FETs.\nRe-check OpenAIO--Whoop / BLHeli hardware before PCB. Consi



DSHOT GPIO14

FET bridge (per phase, Layout A: Ap/Bp/Cp = PWM/high-side, Ac/Bc/Cc = COM/low-side):\nPhase A: Q_PA A03401A (P, C5296722) src=VBAT drain=PHASE_A gate=Ap(P1.0)\n Q_NA A03400A (N, C20917) src=GND drain=PHASE_A gate=Ac(P1.1)\nPhase B: Q_PB/Q_NB same with Bp/Bc (P1.2/P1.3)\nPhase C: Q_PC/Q_NC same with Cp/Cc (P1.4/P1.5)\nBEMF: Am/Bm/Cm (P0.1/0.2/0.3) sense via resistor divider to PHASE_A



U? EFM8BB21 (bee-core lib) – physical pins from EFM8BB21-BC-64QFN20:nLayout A port -> pin:P0.0/Vn=pin2, P0.1/Anc=C/P0.2/Bm=pin20, P0.3/Cm=pin19\nP0.5/RTX=pin17, P1.0/Ap=pin18\nP1.1/Cc=pin13, P1.2/Bg=pin11, P1.3/Bc=pin10\nP1.4/Cp=pin9, P1.5/Cc=pin8. VDD=pin4 (+3V3), GND=pin3+12+EP, RSTB=pin5 (pull-up)\nP2.0/C2D=pin6 (debug). Bypass 100n+4.7u on VDD.

SECTION: ESC – EFM8BB21 + Layout A FET bridge (Bluejay)\nCited: Bluejay Layouts/A.inc – PWM_ACTIVE_HIGH, COM_ACTIVE_HIGH, PWM side HIGH.\nMCU: EFM8BB21F16G–C–QFN20 LCSC C80713. Siik: BLUEJAY LAYOUT A / VBAT MAX 1S.\nDShot: GPIO14 -- Rseries 100–330R --> EFM8 P0.5 (pin 17).\n1S direct–drive P+N whoop topology (no FD6288). Verify Vgs: 3.3V M



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TITLE	Bee Core – ESC EFMB Layout A				
DATE	2026-09-15	REF	Bee Core bring-up – graft sheet	REV	0.1-w
SIZE	A3	DRAWING	via global labels to Minimal RP2350	SHEET	4/5

